

C-DOT

Control-Science Experiment Report

Forecasting foundations • MPC reconciliation • development ablations

**Static remains the production winner.
MPC is not promoted.**

What the evidence says

The presentation claim is deliberately narrower than the development ambition.

SCALE IS PROVEN

12 nodes · 384/384 shards · 85.4 minutes
90.9% aggregate CPU · zero failures · zero swap

FORECAST DATA FOUNDATION

Three 28-day traffic-v2 corpora are complete.
Actual UL/DL rate-bin load replaces nominal labels.

CONTROLLER RESULT

30-pair development: +18.76%
128-seed production: −13.30%
Static remains production winner.

RELEASE DISCIPLINE

No development ablation clears every gate.
Validation and release seeds remain untouched.
Forecast/survival gains are not yet claimed.

PRESENTATION RULE

Use production evidence for production claims.

Use ablations to explain what was learned—not to claim promotion.

Experiment ledger

Every submitted/completed experiment and its initial-plan expectation · 1–7 of 14

COMPLETE	production-evidence-freeze Expected: Freeze and do not regenerate the accepted 12-node production package. 384/384 shards · 85.4 min · 90.9% CPU · zero failures/swap	MATCHED
COMPLETE	mpc-30-vs-128-reconciliation Expected: Resolve the conflicting claims and make the 128-seed production result authoritative. Development +18.76%; production −13.30%; Static authoritative	MATCHED
COMPLETE	traffic-v2-28-day-corpora Expected: Generate train, selection/calibration and untouched-test corpora with actual rate-bin labels. Seeds 46001/2/3 · 80,640 steps each · actual rate-bin labels	MATCHED
PARTIAL	mpc-packed-development-wave Expected: Run independent development ablations on seeds 46101-46112 without touching release seeds. 72 workers · 60 valid pairs · churn attempt failed contract · exit 1	PARTIAL
COMPLETE	churn-trigger-skip-contract-regression Expected: Retained safe policies must be auditable as skipped solves without failing schema validation. Skipped-solve audit status fixed · 22 focused tests pass · rerun 3460	MATCHED
COMPLETE	mpc-development-baseline Expected: Clear every MPC development promotion gate. Mean +11.56% · weighted +3.96% · worst -11.41%	DID NOT MATCH
COMPLETE	mpc-development-adaptive-a0 Expected: Clear every MPC development promotion gate. Mean +1.83% · weighted -3.15% · worst -43.45%	DID NOT MATCH

Full machine-readable metrics, configuration hashes and gate checks: [experiment-ledger.json](#)

Experiment ledger

Every submitted/completed experiment and its initial-plan expectation · 8-14 of 14

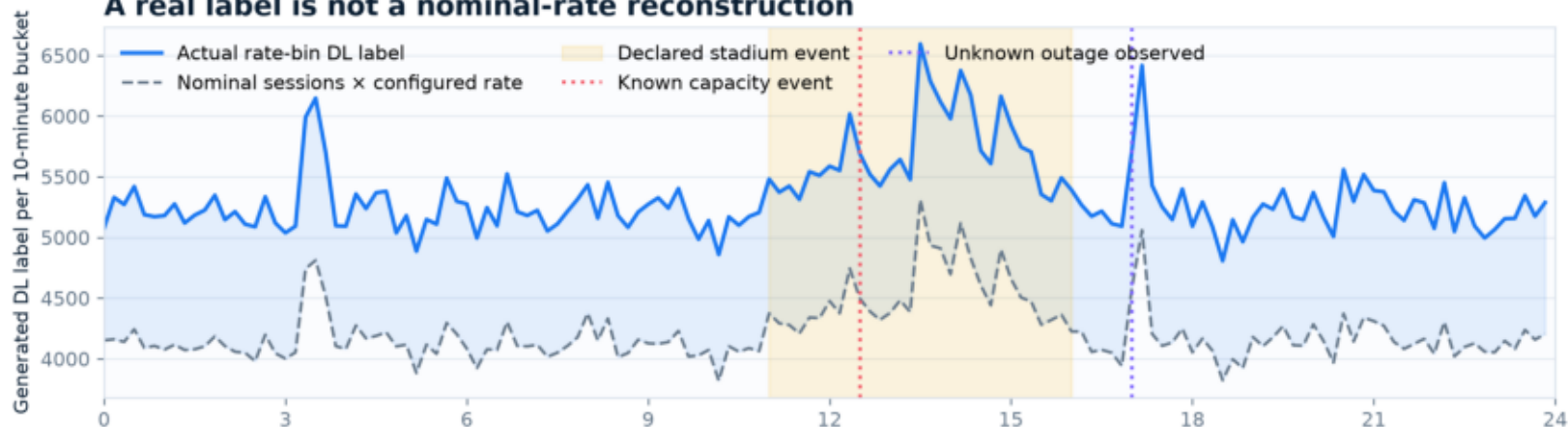
COMPLETE	mpc-development-adaptive-a05 Expected: Clear every MPC development promotion gate. Mean +3.51% · weighted -2.48% · worst -35.31%	DID NOT MATCH
COMPLETE	mpc-development-adaptive-a10 Expected: Clear every MPC development promotion gate. Mean +7.20% · weighted +1.01% · worst -23.82%	DID NOT MATCH
COMPLETE	mpc-development-failure-domain Expected: Clear every MPC development promotion gate. Mean +1.28% · weighted +0.26% · worst -20.12%	DID NOT MATCH
COMPLETE	mpc-development-churn-trigger Expected: Clear every MPC development promotion gate. Mean -22.35% · weighted -28.53% · worst -240.63%	DID NOT MATCH
NOT_RUN	forecast-challenger-comparison Expected: Compare ridge-v2, histogram-gradient quantile, regime ensemble and LightGBM. No outcome claim; intentionally held behind the development gate	PENDING
NOT_RUN	survival-provider-sample-size-and-drift Expected: Compare oracle, empirical, uniform and stale curves at 100/1,000/10,000 samples. No outcome claim; intentionally held behind the development gate	PENDING
NOT_RUN	mpc-validation-and-release Expected: Use validation then untouched release seeds only after development promotion. No outcome claim; intentionally held behind the development gate	CORRECTLY DEFERRED

Full machine-readable metrics, configuration hashes and gate checks: experiment-ledger.json

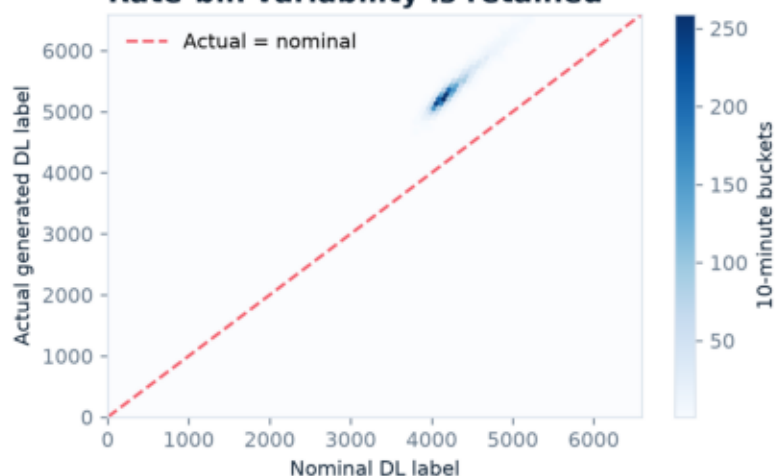
Traffic-v2 realism fingerprint — generated load is causal and rate-bin accurate

Seed 46001 • 28 days • 4,032 decision buckets • median absolute nominal-label gap 26.1%

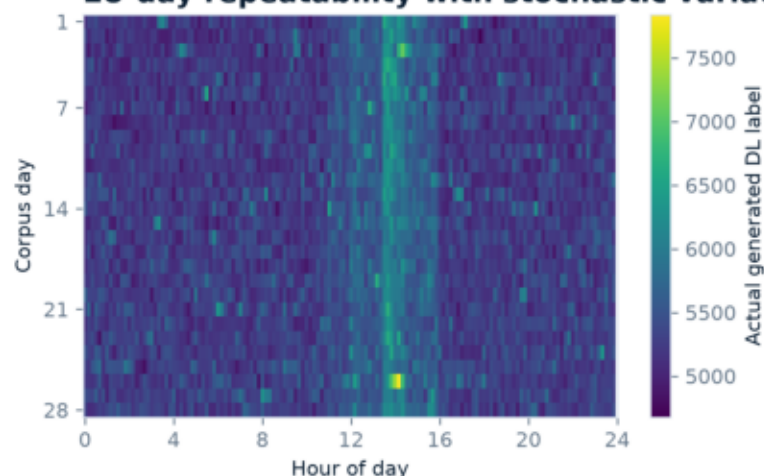
A real label is not a nominal-rate reconstruction



Rate-bin variability is retained

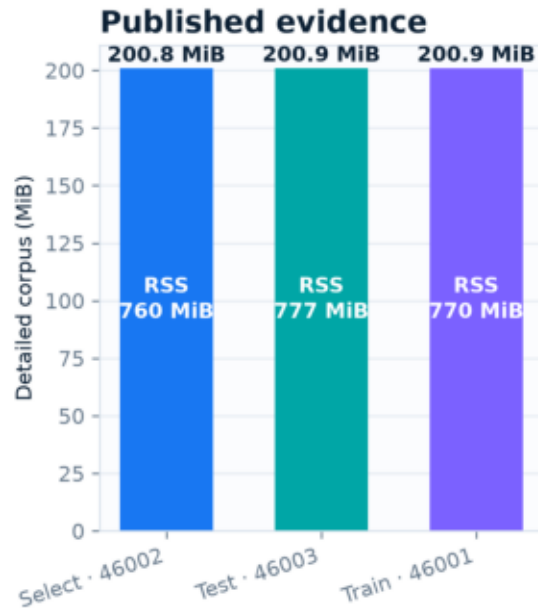
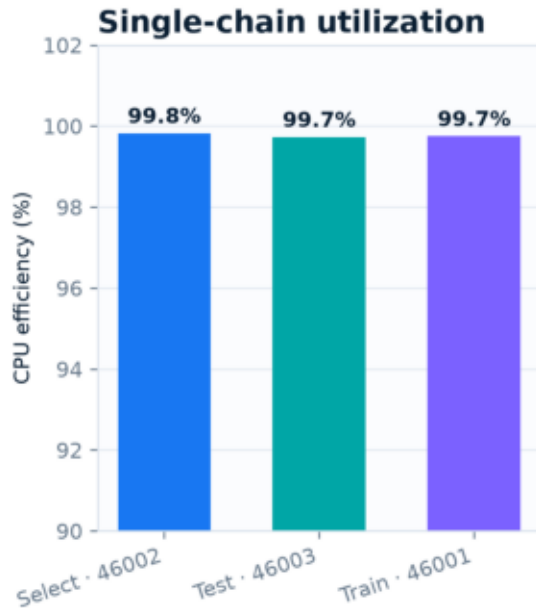
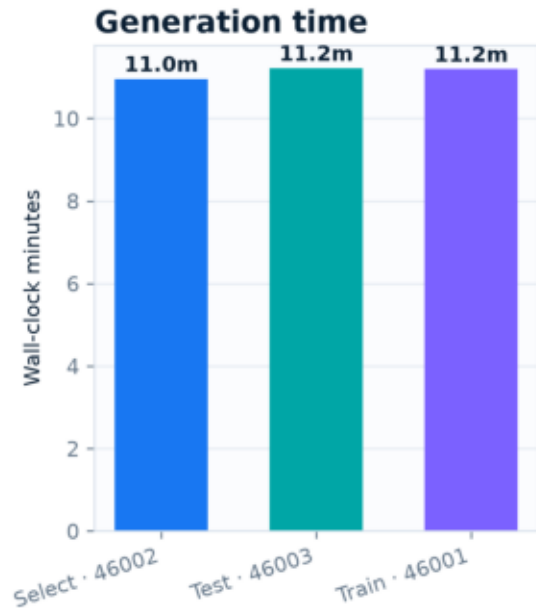


28-day repeatability with stochastic variation



Three untouched 28-day traffic-v2 corpora completed cleanly

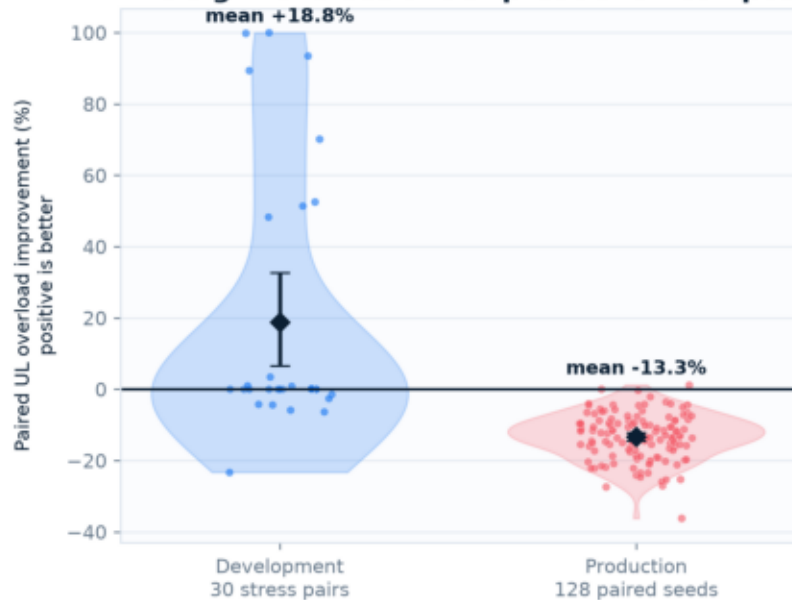
80,640 steps each • actual generated UL/DL labels • zero missing shards



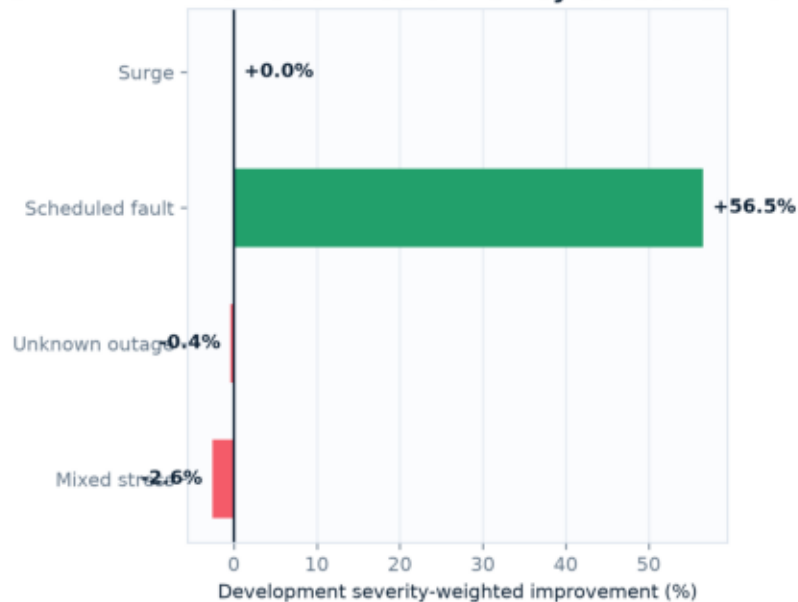
Reconciliation: development promotion $\neq$ production victory

Same metric and model • different scenario contract and seed population • production evidence is authoritative

The sign reverses under production composition



Promotion was driven by scheduled notice

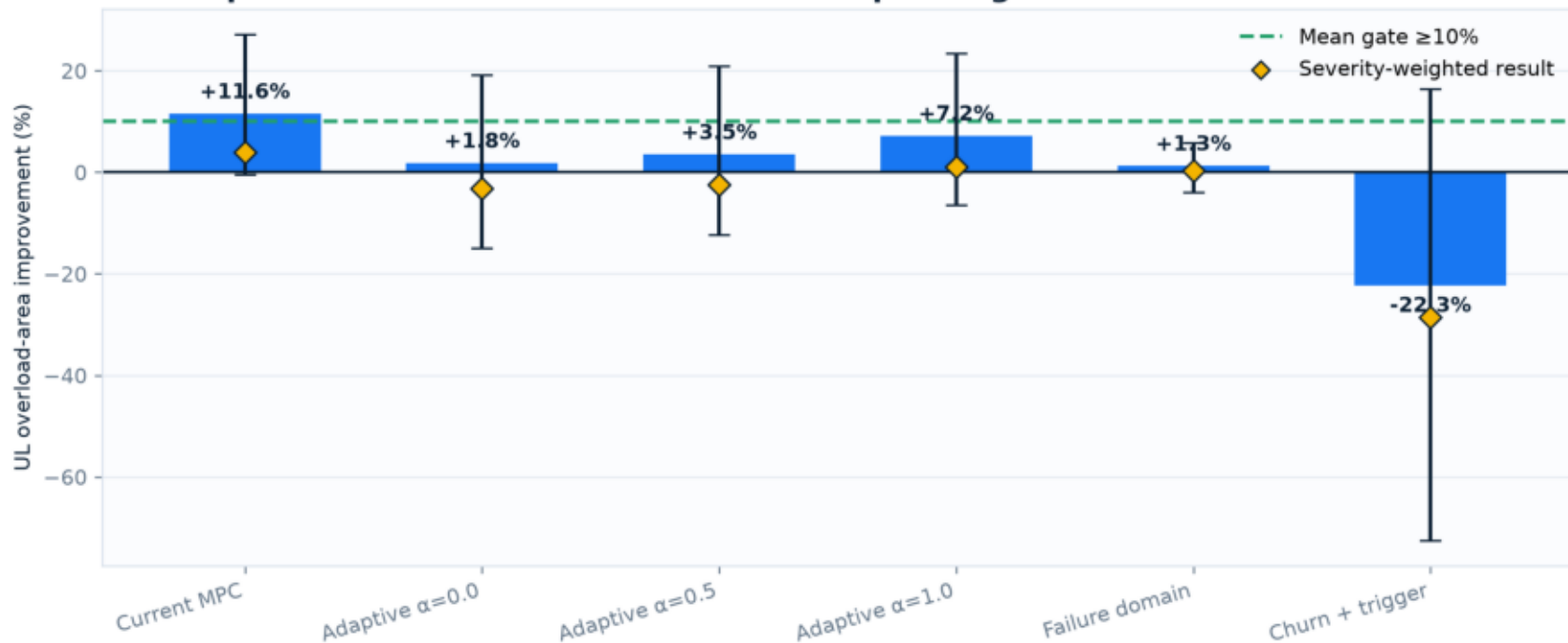


CONCLUSION: STATIC REMAINS PRODUCTION WINNER

MPC development ablation waterfall — paired seeds 46101-46112

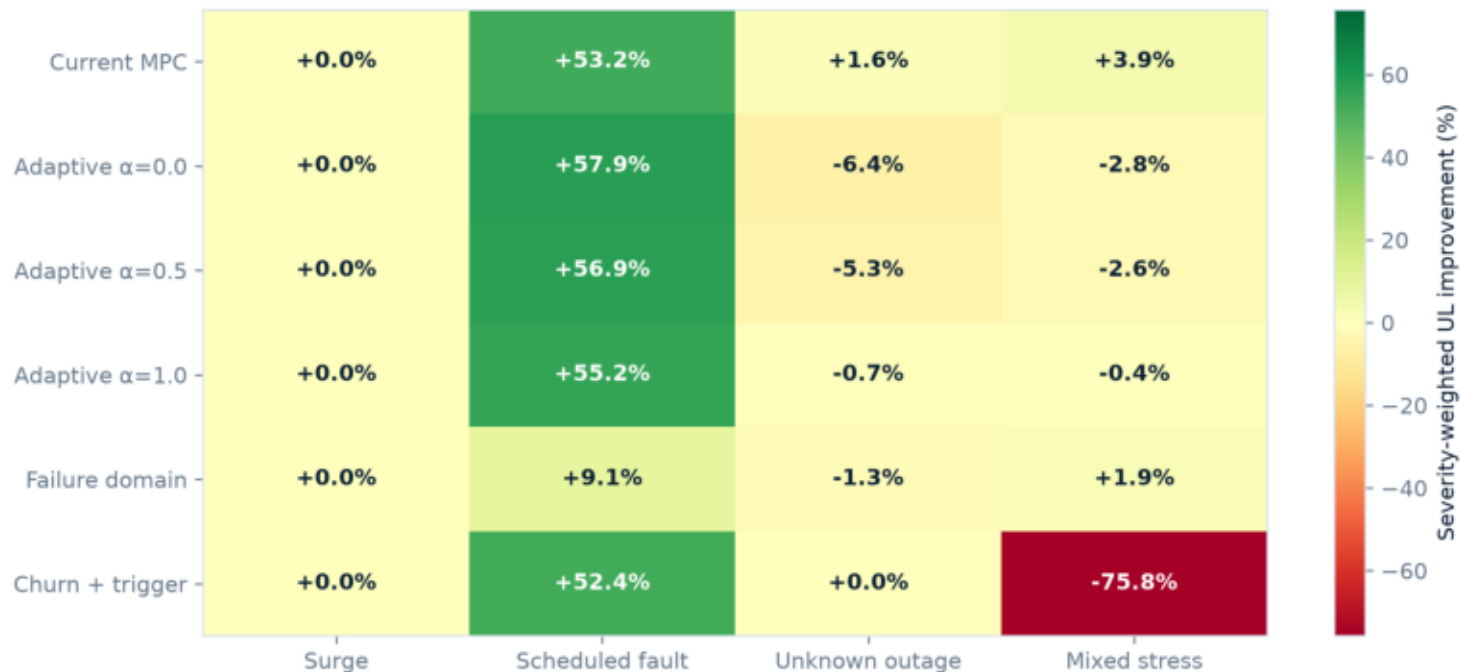
Bars: unweighted pair mean • whiskers: bootstrap 95% CI • diamonds: severity weighted

No completed candidate clears the full development gate

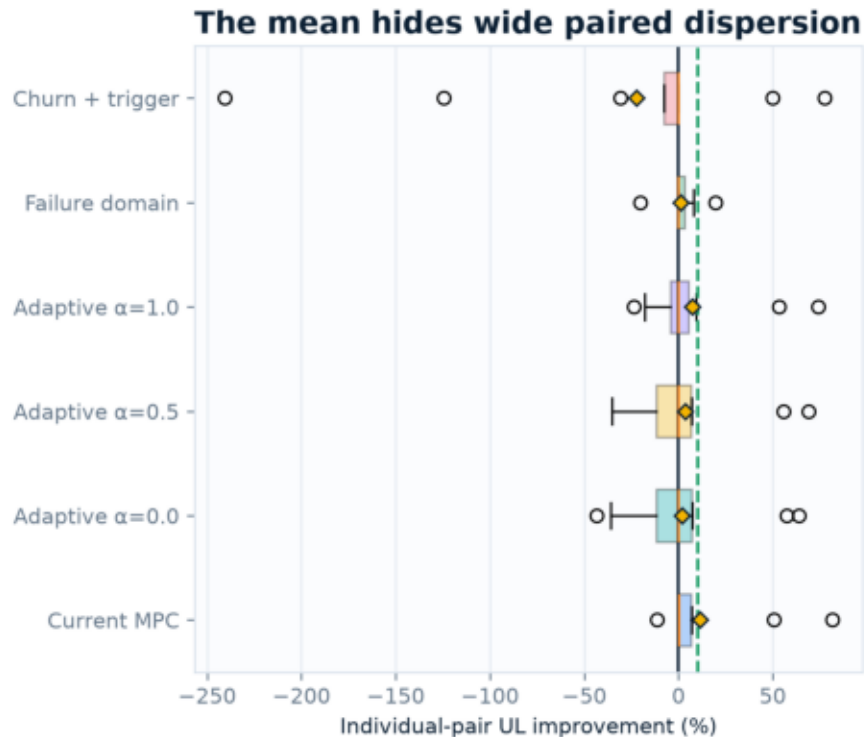
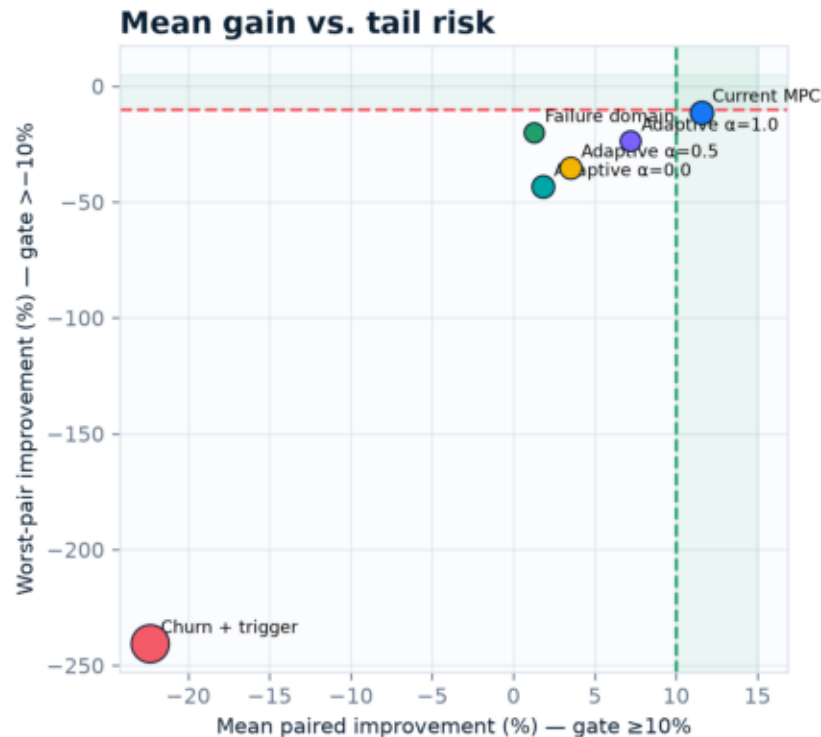


Controllability is scenario-dependent—not a single headline number

Scheduled notice is valuable; mixed stress exposes brittle controller interactions



Release safety requires both average benefit and bounded regressions



Seven-day plan coverage — evidence, not aspiration

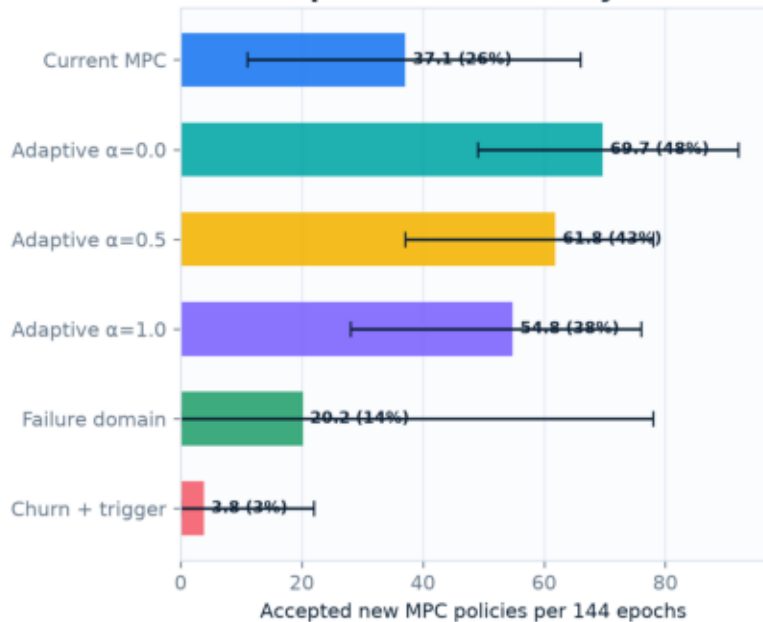
Release and validation seeds remain untouched because no development candidate has cleared the gate

COMPLETE	Frozen 12-node production evidence	384 shards, 90.9% CPU, zero failures
COMPLETE	30-pair vs 128-seed reconciliation	Static remains production winner
COMPLETE	Three 28-day traffic-v2 corpora	Seeds 46001/2/3 published and validated
NOT RUN	Forecast challenger training/test	Promotion criteria not yet testable
NOT RUN	Empirical survival experiments	Estimator implemented; outcome evidence pending
COMPLETE	MPC development ablations	6 of 6 complete; no promotion candidate
COMPLETE	Churn + solve-trigger rerun	12 development pairs after contract fix
NOT RUN	MPC validation seeds 46201-46216	Correctly untouched
NOT RUN	MPC release seeds 46301-46330	Correctly untouched
COMPLETE	New control-science report	Separate from frozen production package

Solve less only matters if overload stays controlled

Whiskers: min-max across paired seeds • retained/skipped policies are excluded • exact L1 route churn was not emitted

Actual optimization activity



Compute reduction vs. network outcome

